

~\OneDrive\Dokument\SKOLA\kiruna\Arne\Arduino\Arne\MPU6050_Lib.cpp

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1  /* MPU_6050_Lib.cpp
2  * Version: 1.1
3  * Created: 2023-04-21
4  * Author: Hjalmar Stenberg & unknown
5  *
6  * Purpose: Library for interaction with MPU6050
7  * System: ESP32 Feather V2
8  * Uses: -
9  *****/
10
11 //NEEDS TO BE UPGRADED
12 #include "MPU6050_Lib.h"
13
14 Adafruit_MPU6050 mpu;
15
16 void initMPU6050(){
17     Serial.println("Adafruit MPU6050 test!");
18
19     // Try to initialize!
20     if (!mpu.begin()) {
21         Serial.println("Failed to find MPU6050 chip");
22         while (1) {
23             delay(10);
24         }
25     }
26     Serial.println("MPU6050 Found!");
27
28     mpu.setAccelerometerRange(MPU6050_RANGE_8_G);
29     Serial.print("Accelerometer range set to: ");
30     switch (mpu.getAccelerometerRange()) {
31     case MPU6050_RANGE_2_G:
32         Serial.println("+2G");
33         break;
34     case MPU6050_RANGE_4_G:
35         Serial.println("+4G");
36         break;
37     case MPU6050_RANGE_8_G:
38         Serial.println("+8G");
39         break;
40     case MPU6050_RANGE_16_G:
41         Serial.println("+16G");
42         break;
43     }
44     mpu.setGyroRange(MPU6050_RANGE_500_DEG);
45     Serial.print("Gyro range set to: ");
46     switch (mpu.getGyroRange()) {
47     case MPU6050_RANGE_250_DEG:
48         Serial.println("+ 250 deg/s");
49         break;
50     case MPU6050_RANGE_500_DEG:
51         Serial.println("+ 500 deg/s");
52         break;
53     case MPU6050_RANGE_1000_DEG:
54         Serial.println("+ 1000 deg/s");
55         break;
56     case MPU6050_RANGE_2000_DEG:
```

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57     Serial.println("+ 2000 deg/s");
58     break;
59 }
60 mpu.setFilterBandwidth(MPU6050_BAND_21_HZ);
61 Serial.print("Filter bandwidth set to: ");
62 switch (mpu.getFilterBandwidth()) {
63     case MPU6050_BAND_260_HZ:
64         Serial.println("260 Hz");
65         break;
66     case MPU6050_BAND_184_HZ:
67         Serial.println("184 Hz");
68         break;
69     case MPU6050_BAND_94_HZ:
70         Serial.println("94 Hz");
71         break;
72     case MPU6050_BAND_44_HZ:
73         Serial.println("44 Hz");
74         break;
75     case MPU6050_BAND_21_HZ:
76         Serial.println("21 Hz");
77         break;
78     case MPU6050_BAND_10_HZ:
79         Serial.println("10 Hz");
80         break;
81     case MPU6050_BAND_5_HZ:
82         Serial.println("5 Hz");
83         break;
84 }
85
86 Serial.println("");
87 delay(100);
88
89 }
90
91 void getReadingsMPU6050(sensors_event_t *a,sensors_event_t *g,sensors_event_t *temp){
92     mpu.getEvent(a, g, temp);
93 }
94

```